Employee Turnover Prediction

Key Findings & Model Performance

Project Objective

Build a machine learning workflow to predict employee turnover, understand turnover patterns, segment employees who left, and support targeted retention strategies.

Project Overview

Dataset	14,999 rows × 10 columns
Rows after duplicate removal	11,991
Missing values	None identified
Selected model	Gradient Boosting

Key Findings

• K-Means clustering identified groups among employees who left based on satisfaction and evaluation.

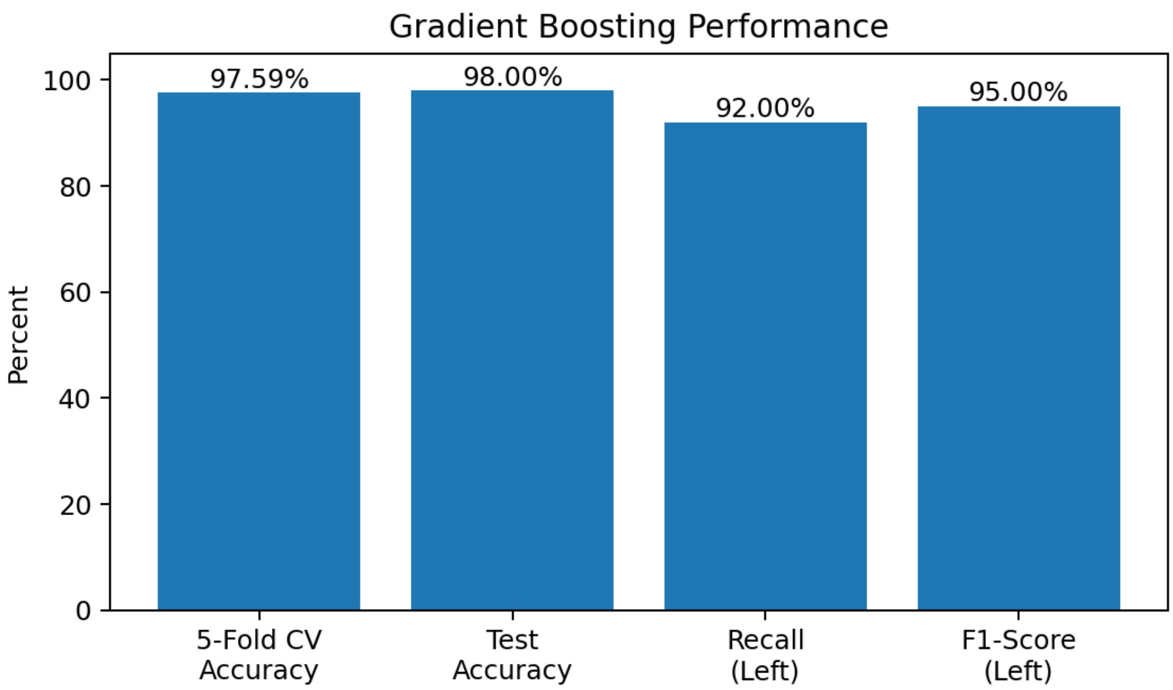
• SMOTE was applied to address class imbalance in the turnover target.

• Logistic Regression, Random Forest, and Gradient Boosting were evaluated using 5-fold cross-validation and classification metrics.

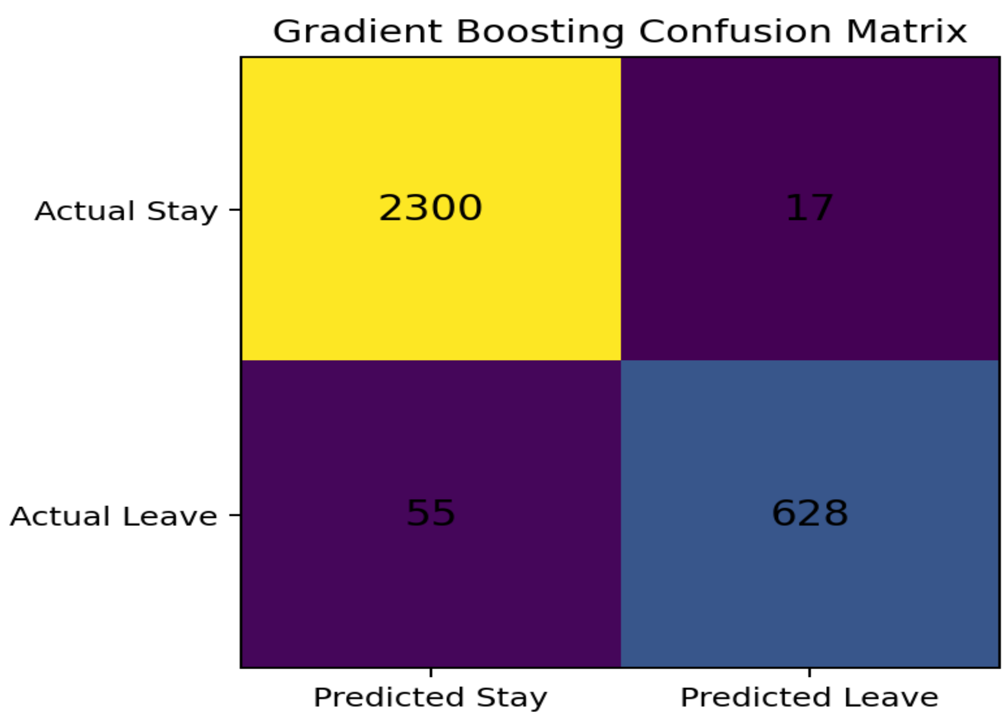
• Gradient Boosting achieved a mean 5-fold cross-validation accuracy of **97.59%**.

• On the test set, Gradient Boosting achieved **98% accuracy**, with **92% recall** and **95% F1-score** for employees who left.

Model Performance



Confusion Matrix



The test-set confusion matrix shows 2,300 correctly predicted employees who stayed, 628 correctly predicted employees who left, 17 false positives, and 55 false negatives.

Employee Risk Segmentation

Predicted turnover probability was used to categorize employees into risk zones for prioritizing retention efforts.

Risk zone	Turnover probability
Safe	<20%
Low Risk	20–60%
Medium Risk	60–90%
High Risk	>90%

Tools & Techniques

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn • Imbalanced-learn • K-Means • SMOTE • Logistic Regression • Random Forest • Gradient Boosting • Cross-Validation